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Failure Modes of Dimensionality Reduction Techniques

A Research Report

Pattern Recognition

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Abstract

Dimensionality reduction is one of the most widely used preprocessing steps in pattern recognition and machine learning. While these techniques are often treated as routine tools, they rest on a range of implicit assumptions that, when violated, can produce serious breakdowns in performance. This report investigates the conditions under which popular dimensionality reduction methods break down, with a focus on Principal Component Analysis (PCA) as the primary case study. The discussion extends to Linear Discriminant Analysis (LDA), Kernel PCA, t-distributed Stochastic Neighbor Embedding (t-SNE), and Uniform Manifold Approximation and Projection (UMAP). Four major categories of breakdown are identified: violations of linearity assumptions, the conflation of variance with discriminative information, sensitivity to noise and outliers, and instability caused by hyperparameter choices in nonlinear methods. For each category, the underlying cause, observable consequences, and possible mitigation are examined, and the findings are summarized in comparison tables across techniques. Overall, the results suggest that no single dimensionality reduction method is universally reliable, and that practitioners must build a critical understanding of method-specific assumptions before applying these tools to real classification tasks.

1. Introduction

High-dimensional data is everywhere. From image recognition to genomics and speech processing, modern pattern recognition systems routinely work with feature vectors that have hundreds or thousands of dimensions. While this richness in representation can be valuable, it also creates serious practical problems. Many learning algorithms degrade in performance as the number of features grows—a phenomenon often referred to as the "curse of dimensionality" [1]. To cope with this, dimensionality reduction has become a standard preprocessing step.

Dimensionality reduction methods aim to map high-dimensional data into a lower-dimensional space while preserving as much of the relevant structure as possible. The best-known of these techniques is Principal Component Analysis (PCA), which has been in use for over a century and remains among the most widely applied methods in both academia and industry [2]. Other techniques, such as LDA, Kernel PCA, t-SNE, and UMAP, have been developed to address different aspects of the problem—particularly when the data structure is nonlinear or when class information is available for supervision.

Despite their widespread use, these techniques are far from perfect. Each method rests on a set of assumptions about the data, and the method tends to work well only as long as those assumptions hold. When they are violated, even partially, the resulting projection can mislead, discard important information, or produce embeddings that actively harm downstream classification. This report refers to such breakdowns as "failure modes."

The goal of this report is to conduct a structured analysis of these failure modes. PCA serves as the primary case study, since it is both the most commonly used and the most thoroughly studied technique, but the discussion is deliberately extended to other methods, both linear and nonlinear, to give a broader picture of where dimensionality reduction tends to go wrong. The reasoning throughout is grounded in theory and supported by established findings in the pattern recognition literature.

One of the reasons for selecting PCA as the main focus of this report is that it remains the most commonly introduced dimensionality reduction technique in pattern recognition courses and practical applications. During the literature review, it became clear that many of the reported failures of PCA also provide useful insight into the limitations of other dimensionality reduction methods. This motivated a broader comparison with LDA, Kernel PCA, t-SNE, and UMAP to better understand when each technique is appropriate.

The remainder of this report is structured as follows. Section 2 reviews related work in the area of dimensionality reduction and its known limitations. Section 3 describes the analytical methodology used in this study. Section 4 presents the main analysis and discussion of failure modes. Section 5 concludes the report and outlines directions for future investigation.

2. Related Work

The foundational reference for PCA remains the 2016 review by Jolliffe and Cadima, published in the Philosophical Transactions of the Royal Society A [2]. The paper gives a thorough treatment of PCA's mathematical basis, its variants, and its known limitations, noting that PCA identifies new uncorrelated variables that successively maximize variance. That same property, though, is also where the method's main weakness comes from: it has no awareness of class labels or of what the downstream task actually requires.

Kernel PCA was introduced by Schölkopf, Smola, and Müller in their 1998 paper in Neural Computation as a way to handle nonlinear structure by mapping data into a high-dimensional feature space through kernel functions [3]. This addressed PCA's linearity limitation, but it came at a price: a suitable kernel has to be chosen for the data, and the method introduces cubic computational complexity with respect to the number of samples.

A major step forward for high-dimensional visualization came with t-SNE, introduced by van der Maaten and Hinton in the Journal of Machine Learning Research [4]. The algorithm modifies its predecessor, SNE, by replacing Gaussian similarity in the low-dimensional space with a Student-t distribution, which helps alleviate the crowding problem—the tendency for moderately distant points to collapse onto each other in the embedding. Even so, t-SNE does not preserve global structure well, and its results depend heavily on the perplexity hyperparameter.

McInnes, Healy, Saul, and Grossberger proposed UMAP in 2018 as a faster and more theoretically grounded alternative to t-SNE [5]. Built on concepts from Riemannian geometry and algebraic topology, UMAP is generally quicker and better at preserving global structure than t-SNE. It still depends, though, on a neighbor parameter that can substantially change how the resulting embedding looks.

A broader review published in recent years covers the full range of dimensionality reduction methods, from PCA through autoencoders, evaluating each in terms of computational complexity, scalability, and how well structure is preserved [6]. It points out that PCA struggles with strong nonlinearities and outliers, while t-SNE and UMAP are good at preserving local structure but distort global relationships. Together, these findings form the theoretical backdrop for the analysis in this report.

Outlier sensitivity in PCA has also been studied from the angle of robust statistics. The sample covariance matrix that PCA relies on is easily distorted by heavy-tailed or contaminated observations, which causes the computed principal directions to drift away from the true structure of the data [7]. This is part of what has motivated robust PCA variants that replace the standard covariance estimator with more resistant alternatives.

3. Methodology

This study takes an analytical approach to investigating failure modes in dimensionality reduction. Rather than designing a new algorithm or running a large-scale benchmark, the goal is to identify and characterize the conditions under which existing methods break down. The objective of this methodology is not to compare algorithms through extensive experimentation, but to understand the theoretical reasons behind their failures. Therefore, emphasis is placed on interpreting published findings, comparing assumptions across techniques, and explaining how these assumptions influence model behavior in different pattern recognition scenarios.

The analysis proceeds in four steps.

First, a review of the theoretical properties of each technique is used to identify the core assumptions that each method relies on. For example, PCA assumes that directions of maximum variance are the most informative, and that the data structure is linear. These assumptions are made explicit.

Second, for each assumption, a corresponding failure condition is described—that is, the data scenario in which the assumption is violated. This ranges from nonlinear manifolds, to imbalanced or noisy data, to overlapping class distributions.

Third, the observable consequences of each failure are discussed—not just the breakdown of the dimensionality reduction step itself, but also its downstream effects on classification performance, clustering quality, or visualization accuracy.

Finally, the analysis is synthesized into structured tables that summarize the failure modes, compare them across techniques, and suggest diagnostic strategies. Datasets are referenced only as illustrative examples and are not the main contribution of the report.

Table 1: Comparison of Dimensionality Reduction Techniques: Methods, Assumptions, and Failure Modes

Technique	Type	Key Assumption	Primary Failure Mode	Scalability
PCA	Linear	Variance = Information	Fails on nonlinear manifolds; sensitive to outliers	High
LDA	Linear (supervised)	Gaussian classes, equal covariance	Breaks when assumptions violated; limited to C-1 dims	High
Kernel PCA	Nonlinear	Correct kernel choice	Kernel selection difficulty; $O(n^3)$ complexity	Low
t-SNE	Nonlinear (local)	Local neighborhoods are meaningful	Distorts global structure; non-reproducible embeddings	Low
UMAP	Nonlinear	Data lies on uniform manifold	Sensitive to hyperparameters; stochastic outputs	Moderate

4. Analysis and Discussion

This section presents the core analytical findings of this study. Each subsection addresses a distinct category of failure mode, explaining the theoretical basis for the failure, the conditions under which it arises, and its practical consequences for pattern recognition tasks.

4.1 Failure Due to Linearity Assumptions

The most basic weakness of PCA is its assumption that the important structure in the data can be captured by linear projections. PCA finds directions in the feature space along which variance is maximized, and when the data genuinely lies along linear axes, this works well. Many real-world datasets, however, have nonlinear structure—data that lies on a curved manifold, for instance, or that forms concentric rings or spirals.

In these situations, PCA produces a projection that cuts through the manifold instead of following it, and the resulting low-dimensional representation loses the nonlinear structure entirely. Classes that were separable in the original space can end up overlapping completely in the projection, which makes the output harmful rather than helpful for classification [2].

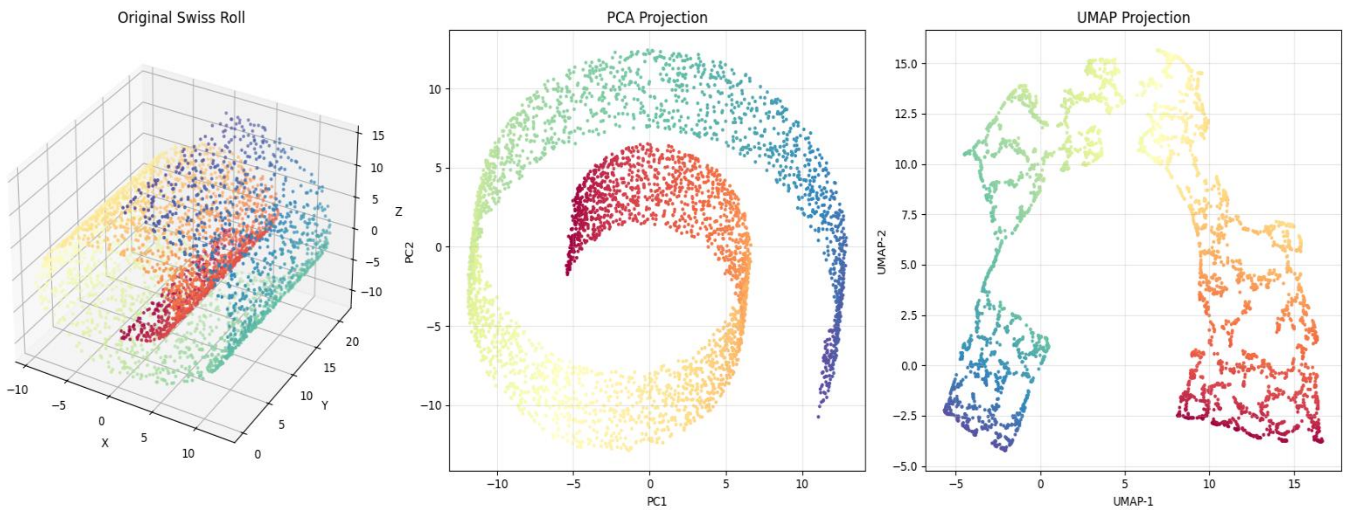

Figure 1. Comparison of PCA and UMAP on the Swiss Roll dataset. PCA collapses the nonlinear manifold into a linear projection, while UMAP preserves the intrinsic neighborhood structure more effectively.

LDA runs into a related but distinct problem. It also produces linear projections, but does so by maximizing the ratio of between-class to within-class scatter. Being a supervised technique gives LDA an edge over PCA for classification tasks, but only when the data satisfies its assumptions: each class is expected to follow a Gaussian distribution, and all classes are assumed to share the same covariance matrix. Once these assumptions are violated, the computed projection directions become suboptimal [6].

Kernel PCA was designed specifically to get around the linearity problem. By mapping data into a high-dimensional reproducing kernel Hilbert space, it can uncover nonlinear principal components in the original input space [3]. That said, this introduces a different weakness: the kernel has to be chosen appropriately for the data. A Gaussian kernel with the wrong bandwidth, or a polynomial kernel of the wrong degree, can produce a feature space poorly suited to the data's geometry. The method is also computationally expensive, scaling as $O(n^3)$ in the number of training samples, which limits how practical it is at scale.

4.2 Variance Is Not Always Discriminative Information

Perhaps the most counterintuitive weakness of PCA is that the directions of maximum variance are not always the directions that best separate the classes. PCA is unsupervised—it has no access to class labels when computing the principal components, and simply maximizes variance under the implicit assumption that informative structure and high variance go together.

That assumption often doesn't hold. Consider a scenario where two classes differ mainly along a direction of relatively low variance, while the dominant variance comes from noise or from a nuisance variable unrelated to class membership. PCA would then discard the discriminative direction when reducing dimensions, producing a projection where the classes overlap even though they were well separated in the original space [2].

Figure 2. PCA Discarding Discriminative Direction

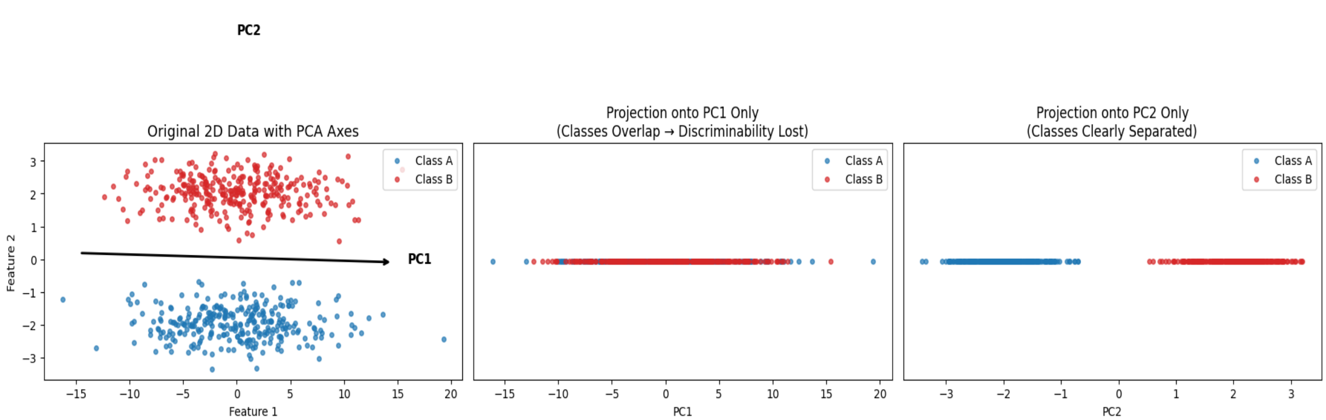

Figure 2. A synthetic two-class dataset where the discriminative information lies along a low-variance direction. Since PCA prioritizes directions with the highest variance, reducing the data to the first principal component may remove the information required for accurate classification.

This is not just a hypothetical concern. It arises whenever a feature has high variance for reasons unrelated to the classification target—measurement noise, irrelevant background variation, or a confounding variable that went uncontrolled during data collection, for example. The resulting projection may look clean and well structured by the criteria PCA optimizes for, such as explained variance, yet it can still destroy the signal needed for accurate classification.

LDA avoids this particular problem because it explicitly uses class label information to compute scatter matrices. It comes with its own assumption-driven weakness, though: if the within-class distributions are not Gaussian, or if the covariance matrices differ substantially across classes, the scatter matrix computations become unreliable and the resulting projection ends up suboptimal [6].

4.3 Sensitivity to Outliers and Noise

Both PCA and LDA are sensitive to outliers, though for different reasons. PCA computes the covariance matrix of the data, and this matrix is heavily influenced by extreme values—a single outlier with extreme coordinates can rotate the principal components away from their true directions, causing the resulting projection to be dominated by noise rather than signal [7]. The sample covariance estimator is known to behave this way; it is simply not robust in a statistical sense.

The effect of outliers is especially severe in high-dimensional data. There is more room for outliers to appear in unusual directions, and the covariance matrix has more entries that can be distorted as a result. Robust PCA research has addressed this by replacing the standard covariance estimator with alternatives that downweight or exclude extreme observations, though these approaches add complexity and are not always available in standard software [7].

LDA is vulnerable in a similar way, since it relies on the mean vectors and scatter matrices of each class, both of which are sensitive to outliers. Even a small fraction of mislabeled samples or extreme feature values can distort the between-class scatter matrix enough to shift the discriminant directions away from their optimal positions.

This issue matters in practice because real-world data is almost never perfectly clean. Measurement errors, data entry mistakes, sensor failures, and adversarial inputs all introduce outliers, and a dimensionality reduction step applied before checking for them can silently carry the problem into every step that follows.

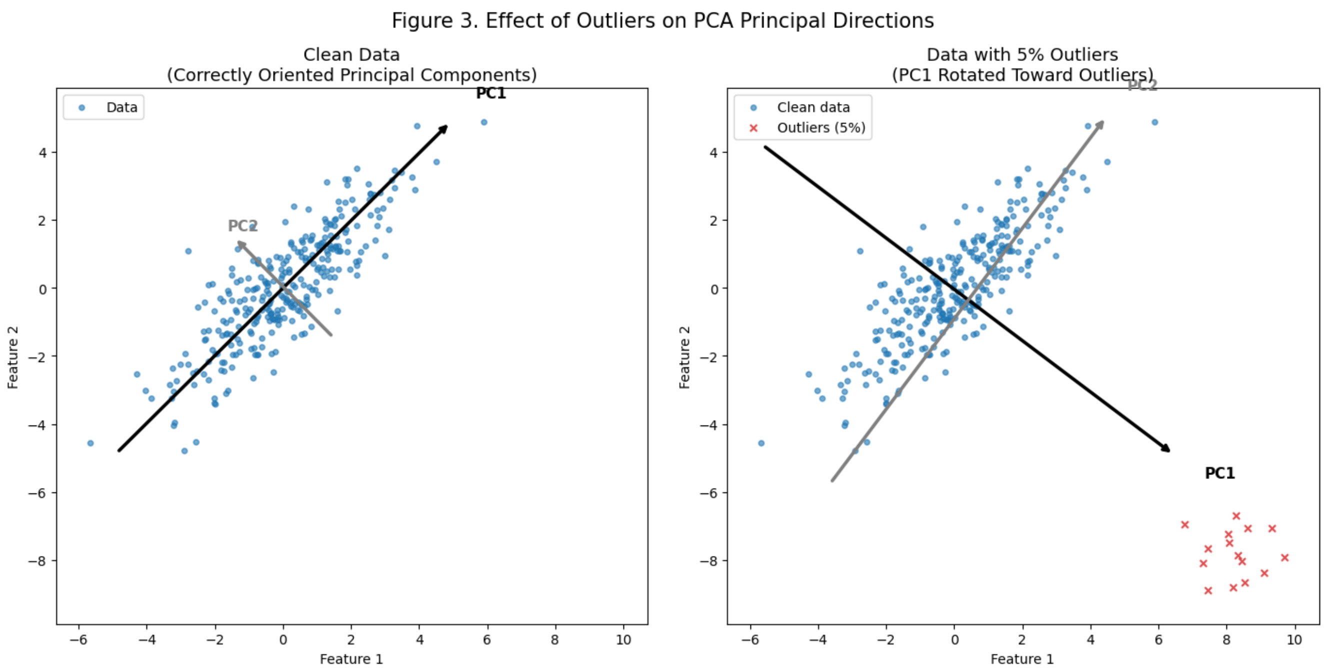

Figure 3. Effect of Outliers on PCA Principal Directions. The plot illustrates how standard PCA is sensitive to extreme values. In the presence of only 5% outliers (right), the first principal component (PC1) rotates significantly away from the true data structure to accommodate the outliers, thereby corrupting the dimensionality reduction process.

4.4 Hyperparameter Sensitivity and Instability in Nonlinear Methods

The nonlinear methods—t-SNE and UMAP—run into a different class of problems centered on their hyperparameters. Both are stochastic optimization algorithms, and both depend on parameters that control the balance between local and global structure in the resulting embedding.

For t-SNE, the key hyperparameter is perplexity, which can be thought of as a soft estimate of how many neighbors each point has. Set it too low, and the algorithm focuses on very local neighborhoods, fragmenting the embedding into many tiny clusters even when the data has continuous structure. Set it too high, and local structure gets over-smoothed, merging distinct clusters together. Van der Maaten and Hinton noted this sensitivity explicitly in the original paper and recommended values between 5 and 50, though following that guidance is difficult without some domain knowledge of the data [4].

More fundamentally, t-SNE does not preserve global structure. The distances between clusters in a t-SNE embedding do not reflect the actual distances between those groups in the high-dimensional space—two clusters that look far apart in a t-SNE plot may actually be close in the original space, and vice versa. This is not a bug so much as a consequence of the algorithm's design: t-SNE's cost function is asymmetric in a way that favors local neighborhood preservation over global layout [4].

UMAP improves on several of these points. It is faster, more scalable, and generally better at preserving aspects of global structure. Even so, it depends on a neighborhood parameter (n_neighbors) and a minimum distance parameter that can substantially change the outcome. UMAP's theoretical foundations assume the data lies on a locally connected, uniformly distributed manifold [5]; when this assumption breaks down—say, when the density of the data varies strongly across the manifold—the resulting embedding can distort inter-cluster relationships in ways that are hard to detect.

Both t-SNE and UMAP also involve a degree of randomness during optimization, meaning that repeated runs can produce slightly different embeddings if the random seed is not fixed. While this does not make the methods unreliable, it highlights the importance of reporting experimental settings and using consistent parameters when results are compared or reproduced in scientific studies.

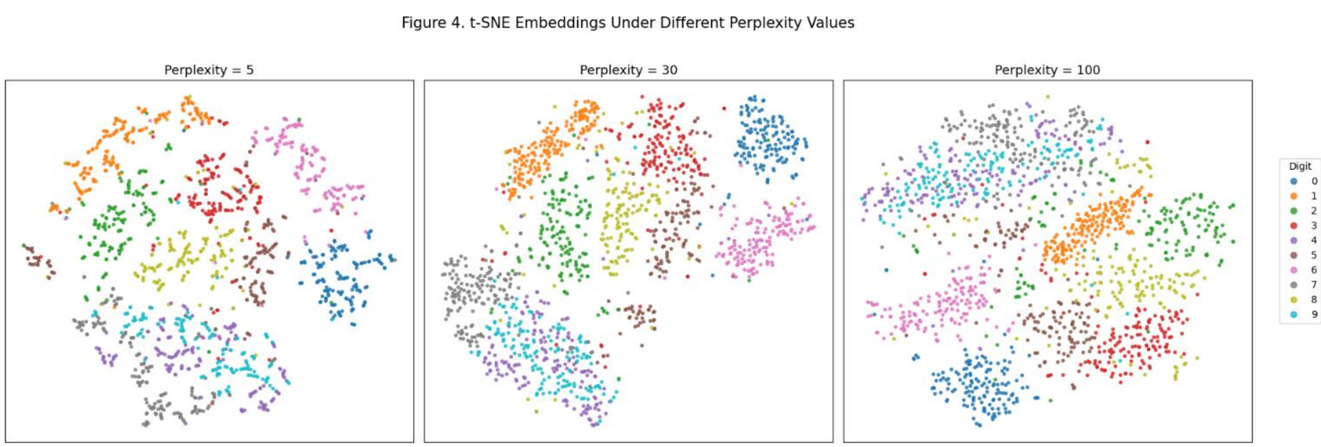

Figure 4. Impact of Perplexity on t-SNE Embeddings. Increasing perplexity shifts the embedding from fragmented local clusters (left) to over-smoothed global structures (right), demonstrating that t-SNE results are highly sensitive to hyperparameter tuning.

4.5 The Problem of Information Loss and Target Dimension Selection

Across all of the methods discussed, a common thread is the problem of information loss. Every dimensionality reduction technique discards some information by definition—the goal is to retain the most important structure while removing redundancy and noise—yet what counts as "important" depends on the downstream task, something these methods do not always have visibility into.

PCA's explained variance criterion is a poor proxy for downstream task performance in many cases. It is possible to retain 95% of the variance in a dataset while losing nearly all of the discriminative information, if the class-relevant signal happens to lie in the low-variance dimensions [2]. This is not a rare edge case; it is a predictable consequence of the method's design.

The degree of information loss also depends on how aggressively the dimensionality is reduced. Choosing the number of components to retain is itself a potential source of error. Common heuristics—such as retaining components that explain at least 1% of variance, or using a scree plot to identify an "elbow"—are useful starting points but are not reliable in all situations. These heuristics say nothing about the relationship between variance and classification performance.

The practical implication is that dimensionality reduction should always be evaluated in the context of the downstream task. A projection that looks good by internal criteria—high explained variance, well-separated clusters in the embedding—may still degrade classifier performance if the wrong information was preserved.

Table 2: Summary of Failure Modes Across Dimensionality Reduction Techniques

Failure Mode	Trigger Condition	Observable Symptom	Affected Techniques
Linearity assumption	Curved or manifold data	Cluster overlap in projection	PCA, LDA
Variance ≠ discriminability	Class separation on low-variance axis	Discarded discriminative features	PCA
Outlier sensitivity	Heavy-tailed or contaminated data	Dominant but uninformative components	PCA, LDA
Kernel misspecification	Wrong kernel for data geometry	Poor nonlinear separation in KPCA	Kernel PCA
Crowding problem	High intrinsic dimensionality	Overlapping points in 2D embedding	t-SNE
Global structure loss	Large-scale cluster relationships	Misleading inter-cluster distances	t-SNE, UMAP
Parameter sensitivity	Poorly tuned perplexity / n_neighbors	Inconsistent or fragmented embeddings	t-SNE, UMAP
Information loss	Aggressive target dimension	Classifier performance degradation	All techniques

Table 3: Diagnostic Strategies for Detecting Dimensionality Reduction Failures

Detection Method	What It Detects	How to Apply
Explained variance plot (scree)	Insufficient variance retention	Check if cumulative variance ≥ 90% before truncating
Reconstruction error	Information loss from projection	Compare original vs. reconstructed data MSE
Downstream classifier accuracy	Loss of discriminative features	Compare classifier AUC before vs. after DR
Stability analysis (multiple runs)	Stochastic inconsistency (t-SNE/UMAP)	Run embedding 10+ times; compare cluster layouts
Silhouette score	Cluster quality in embedded space	Score < 0.5 suggests poor structure preservation

5. Conclusion and Future Work

This report has examined the weaknesses of dimensionality reduction techniques, focusing on the theoretical conditions that lead to breakdowns. Starting from PCA as the central case study and extending the discussion to LDA, Kernel PCA, t-SNE, and UMAP, four main categories were identified: violations of linearity assumptions, the disconnect between variance and discriminability, sensitivity to outliers, and hyperparameter-induced instability in nonlinear methods.

What this analysis shows is that no single dimensionality reduction technique can be trusted in every situation. Each method carries assumptions about the structure of the data, and when those assumptions don't hold, the resulting projection can mislead, destroy useful information, or actively harm downstream classification. Recognizing these failure modes is not just an academic exercise—it is something any pattern recognition practitioner needs to be able to do in practice.

A few directions for future work follow from this. There is a need for better diagnostic tools that can catch dimensionality reduction problems before they propagate into downstream models. Developing robust variants—particularly PCA methods that resist outliers and do not require careful kernel tuning—also remains an active and important research area. And the question of how to choose the number of dimensions to retain, guided by downstream task performance rather than variance alone, deserves more systematic attention.

More broadly, this analysis points to something important about the role of assumptions in pattern recognition. Techniques work when their assumptions hold and break down when they don't. Learning to recognize when those assumptions are at risk—and choosing methods accordingly—is one of the most valuable skills a pattern recognition practitioner can develop.

While preparing this report, I found that dimensionality reduction should not be viewed as a routine preprocessing step. Instead, selecting an appropriate technique requires understanding the characteristics of the data and the assumptions behind each method. This analytical perspective helped me better appreciate how theoretical concepts directly influence the performance of pattern recognition systems.

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